

BRANDON MINER

San Francisco, CA

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AI Engineer — Multi-modal pipelines, LLM-assisted labeling, and production inference for unstructured video and text

Education

University of San Francisco

M.S. in Data Science and Artificial Intelligence

San Francisco, CA

Expected Jun 2026

San Diego State University

B.S. in Statistics with Emphasis in Data Science, Minor in Mathematics

San Diego, CA

Dec 2025

Experience

AI/ML Engineer

ACLU

San Francisco, CA

Oct 2025 – Present

- Architecting the software design of an end-to-end **multi-modal AI** video auditing platform from scratch (**Python, FastAPI, MySQL, Docker**) to automate analysis of unstructured police bodycam footage, projected to eliminate **300+ hours** of manual review per monthly cycle, all while tracking product development.
- Engineered a **GPU-accelerated inference pipeline** using **FFmpeg** and **OpenAI Whisper** for speech-to-text transcription, processing **600+ videos per month**; integrated **LLM-driven auto-labeling** to generate training data for fine-tuning multi-stage text classification models.
- Architected and shipped a **multi-stage NLP classification system** combining LLM-assisted auto-labeling, iterative fine-tuning, and structured output parsing — delivering audit-ready signals directly in the hands of end-users.
- Designed and deployed a **private-server FastAPI backend** paired with a **React.js** frontend, delivering the full AI auditing system as a production-ready web application for non-technical end-users.

Data Scientist, Lead Intern

San Diego County Taxpayers Association

San Diego, CA

Mar 2024 – Aug 2025

- Led a team of **4 analysts** to build data pipelines delivering statistical analysis of civil workforce compensation and homelessness intervention programs.
- Developed an **data standardization pipeline** combining iterative web scraping and text normalization to unify messy public wage records; applied regression modeling to quantify a **19.7% police wage increase** per **1%** staffing reduction.
- Applied **mixed-effects regression models** and **multicollinearity reduction** to analyze regional homelessness expenditure data, identifying high-impact interventions (e.g., rapid rehousing) to guide future resource allocation decisions.

Technical Projects

GenAI Franchise Personality Quiz

- Built an end-to-end quiz generator: users describe any fictional class system in natural language; Gemini parses the franchise, scrapes Fandom wikis for canon characters, and roleplays each through 15 Likert questions to produce reference vectors asynchronously via FastAPI background jobs.
- Implemented a vector-based classifier that scores user answers with cosine similarity against LLM-generated character embeddings, ranks classes by per-type mean similarity, and projects results into 3D PCA for interactive visualization in a Next.js results UI.
- Deployed a split architecture on Vercel (frontend) and Google Cloud Run (API), with durable quiz artifacts in GCS, shareable quiz URLs, a community quiz library, and GitHub Actions CI (pre-commit, pytest, automated API deploy).

Parkcast-SF: San Francisco Parking Assistant

- Owned system design and the production **FastAPI** inference API for a full-stack USF capstone parking platform, serving block-level **LightGBM** occupancy predictions from artifacts in **GCS**.
- Applied **K-nearest neighbors**-derived baselines to support **LightGBM** inference on sparse, non-metered blocks where direct sensor history was unavailable.
- Built automated model **CI/CD** on **GCP**: scheduled **GitHub Actions** retraining with **MLflow** experiment tracking and a **champion/challenger** promotion gate (validation MAE) before GCS upload; split deploy across **Vercel** (frontend) and **Cloud Run** (API).

Technical Skills

Programming & Frameworks: Python (PyTorch, Transformers, Hugging Face, FastAPI, Pandas, NumPy, Scikit-learn), SQL (PostgreSQL), JavaScript (React.js), Bash

AI / ML: LLM integration & prompting, RAG/agent orchestration, multi-modal AI, NLP (ASR, Whisper, text classification, fine-tuning), GPU-accelerated inference, ROC-AUC threshold optimization, PCA

MLOps & Deployment: Docker, Docker Compose, Google Cloud Run, REST APIs (FastAPI), private-server deployment, GitHub Actions CI/CD, Git

Data: multi-source ETL pipelines, web scraping, entity standardization